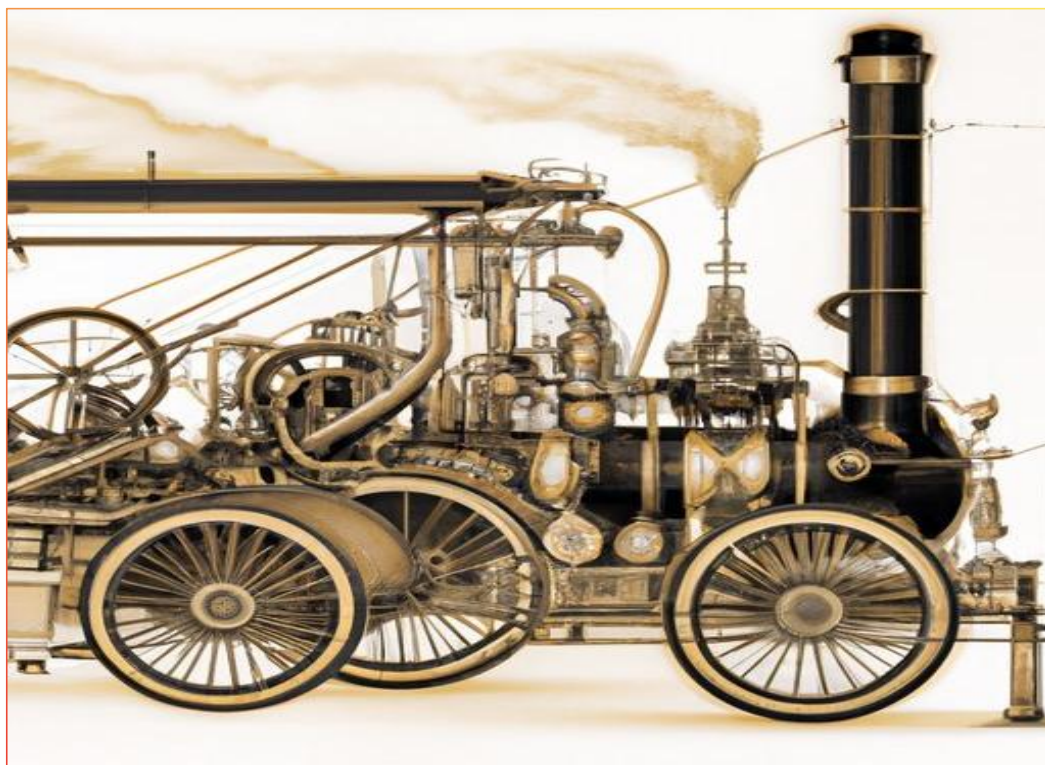


Unit 3

Steam Engine



Looking Back

- 🔥 **Reading 1:**
Who Invented the Steam Engine
- 🔥 **Reading Comprehension**
- 🔥 **Skills:** Cohesion



Looking Forward

- 🔥 **Reading 2:**
What the Steam Engine Can Tell
Us About the Future of Social
Data
- 🔥 **Translation**
- 🔥 **Further Discussion**



Looking Back

Who Invented the Steam Engine¹

By Elizabeth Peterson

1. In a world propelled by combustion engines, gas turbines and nuclear reactors, the steam engine may seem like a relic of the past. But without this game-changing invention, the modern world would be a much different place.

2. Arguably the most important development of the Industrial Revolution, the steam engine facilitated major advancements in the fields of mining, manufacturing, agriculture and transportation. And while several prominent figures of the 18th and 19th centuries are credited with developing and improving the steam engine, the history of steam-powered machines actually goes back nearly 2,000 years before the Industrial Revolution.

Ancient steam turbines

3. Early in the first century A.D., a Greek inventor named Heron of Alexandria designed the world's first aeolipile, or primitive steam turbine. Heron's aeolipile consisted of a hollow sphere, mounted on a pair of tubes. Heated from below by fire, the tubes transported steam to the sphere, where it was released through another series of tubes projecting from the sphere's equator. This movement of steam through the device caused the sphere to revolve, demonstrating the potential for using steam as a means of propulsion.

¹ The original article is available at <https://www.livescience.com/44186-who-invented-the-steam-engine.html>

Unit 3 Steam Engine—Looking Back



(In the first century A.D., Heron of Alexandria invented the aeolipile, or primitive steam turbine.)

4. While Heron's aeolipile was created as a novelty, not a means of speeding up production, it is nevertheless the first known device to transform steam into rotary motion. But it wasn't until the 17th century that attempts were made to harness the power exhibited by Heron's aeolipile for practical purposes.

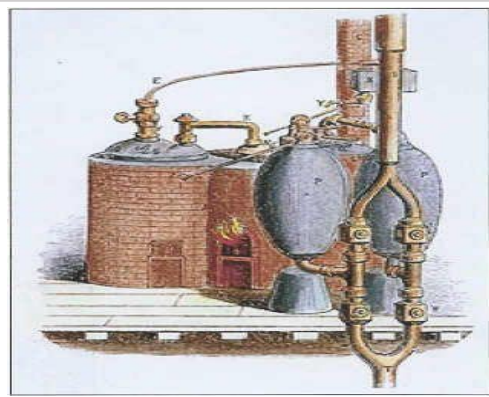
Steam: A perfect solution

5. The first practical steam engines were developed to solve a very specific problem: how to remove water from flooded mines. As Europeans of the 17th century switched from wood to coal as their main source of fuel, mines were deepened and, as a result, often became flooded after penetrating underground water sources.

6. A Spanish mining administrator named Jerónimo de Ayanz is thought to have been the first person to solve the problem of flooded mines. In 1606, de Ayanz registered the first patent for a machine that used steam power to propel water from mines. The Spanish inventor — who is also credited with inventing one of the world's first air conditioning systems — used his steam engine to remove water from silver mines in Guadalcanal, Seville.

Unit 3 Steam Engine—Looking Back

7. While the Spaniard first patented a steam-operated machine for use in mining, an Englishman is usually credited with inventing the first steam engine. In 1698, Thomas Savery, an engineer and inventor, patented a machine that could effectively draw water from flooded mines using steam pressure. Savery used principles set forth by Denis Papin, a French-born British physicist who invented the



(In 1698, Thomas Savery patented a machine that could effectively draw water from flooded mines using steam pressure.)

pressure cooker. Papin's ideas surrounding a cylinder and piston steam engine had not previously been used to build a working engine, but by 1705, Savery had turned Papin's ideas into a useful invention.

8. Using two steam boilers, Savery devised a nearly continuous system for pumping water from mines. But despite the early success of Savery's system, it was soon discovered that his engine was only capable of drawing water from shallow depths, a problem that needed to be overcome if steam engines were to function in deep mines.

9. Luckily for European mine owners, in 1711 another Englishman, Thomas Newcomen, developed a better way to pump water from mines. His system used a redesigned steam engine that eliminated the need for accumulated steam pressure — a flaw in Savery's system that led to many an unfortunate explosion. Newcomen's "atmospheric" engine — so named because the level of steam pressure it used neared atmospheric pressure — was the first commercially successful machine that used steam to operate a water pump.

10. Despite it's being an improvement on Savery's initial rendering of the steam engine, Newcomen's atmospheric engine also had its flaws. The machine was highly inefficient, requiring a constant flow of cold water to cool the all-important steam cylinder (the part of the engine where steam pressure is converted into

Unit 3 Steam Engine—Looking Back

motion), as well as a constant energy source to reheat the cylinder.

11. Regardless of this major drawback, Newcomen's engine design went unchallenged for the next 50-or-so years and, aside from pumping out mines, was also used to drain wetlands, supply water to towns and even power factories and mills by pumping water from below a water wheel to above it for re-use.

Powering the Industrial Revolution

12. But by 1765, the fate of Newcomen's engine was sealed. In that year, James Watt, a Scottish instrument maker employed by Glasgow University, began repairing a small model of a Newcomen engine. Watt was perplexed by the large amount of steam consumed by Newcomen's machine and realized that to remedy this inefficiency, he would have to do away with the constant cooling and reheating of the steam cylinder.

13. To do this, Watt developed a separate condenser, which allowed the steam cylinder to be maintained at a constant temperature and dramatically improved the functionality of Newcomen's engine.

14. For financial reasons, Watt wasn't immediately able to manufacture his new and improved atmospheric engine. But by 1776, he had formed a partnership with Matthew Boulton, an English manufacturer and engineer dead-set on using steam engines for more than just pumping water from mines.

15. With financial backing from Boulton, Watt developed a single-acting (and later, a double-acting) rotative steam engine that, along with Watt's signature separate condenser, featured a parallel motion mechanism that doubled the power of the existing steam cylinder. The Boulton-Watt engine was also the first that allowed the machine's operator to control the engine speed with a device called a centrifugal governor. The improved engine used a new gear system — developed by Boulton and Watts' employee, William Murdoch — known as sun and planet

Unit 3 Steam Engine—Looking Back

gearing, to convert reciprocating (linear) motion into rotative motion.

16. Watt's improvements to the steam engine, combined with Boulton's vision of a nation powered by steam, facilitated the rapid adoption of steam engines across the United Kingdom and, eventually, the United States. By the 1800s, steam engines were powering mills, factories, breweries and a host of other manufacturing operations. In 1852, the first flight of a steam-powered airship took place. Future iterations of the steam engine also came to define travel, as trains, boats and railways adopted the technology to propel passengers into the 20th century.

Reading Comprehension

Analyzing and Extracting

I. Choose the appropriate letters A-D for each question.

1. According to the passage, what role did the steam engine play in the Industrial Revolution? _____

- A. It was a minor development with limited impact.
- B. It promoted progress in many fields such as mining.
- C. It had a significant impact mainly in the field of agriculture.
- D. It was primarily used for entertainment purposes.

2. What function did Heron's aeolipile serve when it was first created? _____

- A. It was used to speed up industrial production.
- B. It was an early form of heating system.
- C. It demonstrated the potential for using steam to create motion.
- D. It was used to pump water from mines.

3. What significant improvement did James Watt make to Newcomen's steam engine? _____

- A. He increased the size of the steam cylinder.
- B. He developed a separate condenser to maintain a constant temperature.
- C. He added more steam to the engine.
- D. He changed the fuel source from coal to wood.

I. Do the following statements agree with the views of the writer in the reading passage?

Unit 3 Steam Engine—Looking Back

True	if the statement agrees with the writer
False	if the statement does not agree with the writer
NOT GIVEN	if there is no information about tills in the passage

4. _____ Jerónimo de Ayanz used his steam engine to remove water from silver mines.
5. _____ Thomas Savery's steam engine was widely adopted for use in deep mines immediately after its invention.
6. _____ Thomas Newcomen developed a better way to pump water from mines by eliminating the need for accumulated steam pressure.
7. _____ Newcomen's atmospheric engine was capable of maintaining a constant temperature in the steam cylinder.
8. _____ James Watt was employed as an engineer by Matthew Boulton before the two formed a partnership.

Skills

Cohesion

1. What is cohesion?

- ✧ Cohesion refers to the **grammatical** and **lexical** relationships between words and phrases.
- ✧ It is the glue that holds sentences together, making them easy to read and understand.

2. Types of cohesive devices:

- ✧ Repetitions: to repeat words or ideas using different words (synonyms)
- ✧ Reference words: to refer to something mentioned elsewhere in the text, esp. pronouns, anaphoric nouns
- ✧ Conjunctions: to show the relationship between ideas with linking words, like for example, in contrast, first, as a result...
- ✧ Substitutions: to replace one thing with another for the sake of economy and naturalness, e.g. one, do, the same, so...
- ✧ Ellipsis: to omit a word or phrase when the meaning can be retrieved from earlier in the text, also called substitution by zero

Sample for Cohesion:

Q: Why did Europeans of the 17th century switch their main source of fuel from wood to coal?

In the 17th century, Europeans switched their primary fuel source from wood to coal for a few good reasons. Firstly, they were running low on wood due to the depletion of this resource, so they needed a backup plan. Moreover, coal proved to be a more reliable source of energy. It was not just more efficient but could burn at higher temperatures, which was essential for new industries. Additionally, thanks to advances in mining technologies and transportation infrastructure, such as canals, coal became cheaper and easier to get, allowing industries to increase their profit margin. Therefore, the switch was not merely a choice — it was a must-do, reshaping Europe's energy scene for the better.



Select a topic from the list and deliver a one-minute presentation.

1. The reasons why it took so long for steam power to be used in industry since Heron's invention of Aeoliple

2. The pre-conditions for the invention of the steam engine



Looking Forward

What the Steam Engine Can Tell Us About the Future of Social Data²

By Mark Cartwright

1. We know that the spread of computers into everyday life has changed the world around us in dramatic ways. The metaphor people often use is that we're going through another Industrial Revolution. In the same way that the steam engine upended the economy of the 19th century, so too will the proliferation of social data change the economy of the 21st century.
2. We used to think that the Industrial Revolution was a sudden explosion of growth and invention—all at once, a “wave of gadgets” sweeping over Britain in the space of a few years. But that's not true. Economic historians found that there was no “take off” in the British economy; instead, the economy grew slowly and steadily for the century or so before the Industrial Revolution. We get the same story when we look at the technology of the ‘leading sectors’ of the Industrial Revolution. There wasn't a single genius invention that changed everything; instead, the slow iteration of inventions over generations that over time aggregated to an unrecognizable world.
3. The Newcomen engine was not a dead-end curiosity, however. It found great practical utility in coal mining. Coal mining was a big business in those days because people throughout Britain relied on coal to heat their houses. People who owned coal mines needed to get coal out of the ground so they could sell it to these cold consumers. Luckily, there was a lot of easily accessible coal in Britain. But when miners dug deep enough, they would hit the water table, flooding the mine. Miners needed to invest in expensive pumping machinery to drain deeper and deeper mines. Before the Newcomen engine, this was done by people or horses

² The original article is available at <https://mackieland.medium.com/what-the-steam-engine-can-tell-us-about-the-future-of-social-data-16447331e63e>

Unit 3 Steam Engine—Looking Forward

working a treadmill, which was obviously expensive and difficult, limiting the depth to which mines could be sunk. The Newcomen engine revolutionized mine drainage, allowing mine owners to dig deeper and deeper mines, keeping British houses warm in winter and British skies smoky all year round.

4. The vast bulk of coal's cost came from transportation. At the pithead, its cost was close to zero. This was what made the Newcomen engine economic: it didn't matter that the Newcomen engine was incredibly fuel-hungry because fuel was close to free. People didn't have much incentive to improve its efficiency either—why spend money and time tinkering with your engine when you'd be saving basically no money at all from your efficiency gains?

5. It would take a number of other innovations to make the steam engine change the world: first, those made by James Watt and Matthew Bolton that allowed more efficient steam power to move in circles and thus power machinery, and later, the development of much more efficient high-pressure steam engines that would lead to innovations like the steam locomotive. These cumulative innovations were important in that by making the steam engine more efficient; they allowed it to be used far away from the point of coal production. They meant that energy from coal could be used anywhere—not just at the coal pit—but in factories, in houses, even in the deep ocean. It was this development that made the Industrial Revolution when steam power could be used far away from energy sources.

6. When we analogize between our time and the Industrial Revolution, we often think that the process has been completed. When we look around at the changes that have happened over our lifetimes—laptops, smartphones, cloud computing, remote work—we're looking at the end of the story. But the lesson of the slow Industrial Revolution is that we're wrong.

7. We're not at the end of the story. It's more like we're at the time of Newcomen—or, more likely, the time of Watt. We are still using data primarily at the site of production, where it is extremely cheap and ubiquitous. We are still stuck at the metaphorical pithead: our data is still coming from interactions mainly on

Unit 3 Steam Engine—Looking Forward

screens, in apps, with immaterial stuff. The real revolution—the real change—is still to come, when the cost of gathering data decreases to the point where it spreads to more and more domains—where we figure out how to move social data away from our screens and out into the world.

8. How will this happen? It's useful here to see what happened to the steam engine after Watt. James Watt was an inventive genius. He took the Newcomen engine and, over the course of his career, improved it in just about every way imaginable. For his pains, he had a series of patents on his inventions, which allowed him and his business partners a comfortable life.

9. But Watt, for all of his genius, had his untested assumptions that held back the development of the steam engine. One of his assumptions was that it was really dangerous to experiment with high-pressure steam engines. He had some experience with this: high-pressure engines could explode, destroying workshops and potentially killing people. Because of his own anxiety about the risk of experimenting with high-pressure steam, Watt refused to do so. And because he sat on all the patents, nobody else could work with high-pressure steam and validate his assumptions.

10. In the early 19th century, the engineer Richard Trevithick started to experiment with high-pressure steam in Cornwall. Cornwall had no coalfields, so coal was very expensive, which meant Cornish engineers had more cause to seek efficient energy use. What Trevithick found was that high pressure was not particularly dangerous—and it was much more efficient and could make the engines themselves much smaller. It was this reduction in size that made steam-powered transport possible. But it would only take other engineers after the death of Watt and the lapse of his patents to really deliver on the power of high-pressure steam and make the first steam-powered locomotives.

11. Look at the situation today. The good news is that there's a lot of space for innovators and entrepreneurs to push marginal improvements in efficiency in the collection and analysis of social data to new business problems. There will remain

Unit 3 Steam Engine—Looking Forward

an opportunity for a long time—a great opportunity—to completely upend the way that we live, work, and play—because all of these domains will likely be upended by the expansion of social data.

12. This change will likely come by violating the assumptions and boundaries that many of the people in power today hold sacred. Just as the steam engine could only develop when people were able to try to invalidate the assumptions of Watt and others that high-pressure steam was dangerous, there will be a new generation of disruptors who may make advances by violating strongly held norms. These may be about privacy, equity, or the limits of social control. But they will be tested—and many of them will fall to the wayside.

13. The bad news is that the big change will lead to a great deal of individual pain and social upheaval. The geniuses of the Industrial Revolution were great at figuring out their technology, but they were insensitive to the massive social problems that their machines caused. They believed—certainly wrongly—that the people who technology left behind deserved their fate, and they could simply improve their lot through hard work. They believed—perhaps correctly—that the good they did by advancing technology was worth the pain it had caused.

14. The next generation will experience completely new problems because of the proliferation of social data. In the Industrial Revolution, the moral compass of inventors and business owners failed: they allowed their new machines to chew through generations of workers, and they allowed the benefits of this prosperity to become greatly unevenly distributed. The next generation of leaders and thinkers is going to have to confront a much bigger challenge, and it is unclear whether we have the moral judgement to do so well.

Translation

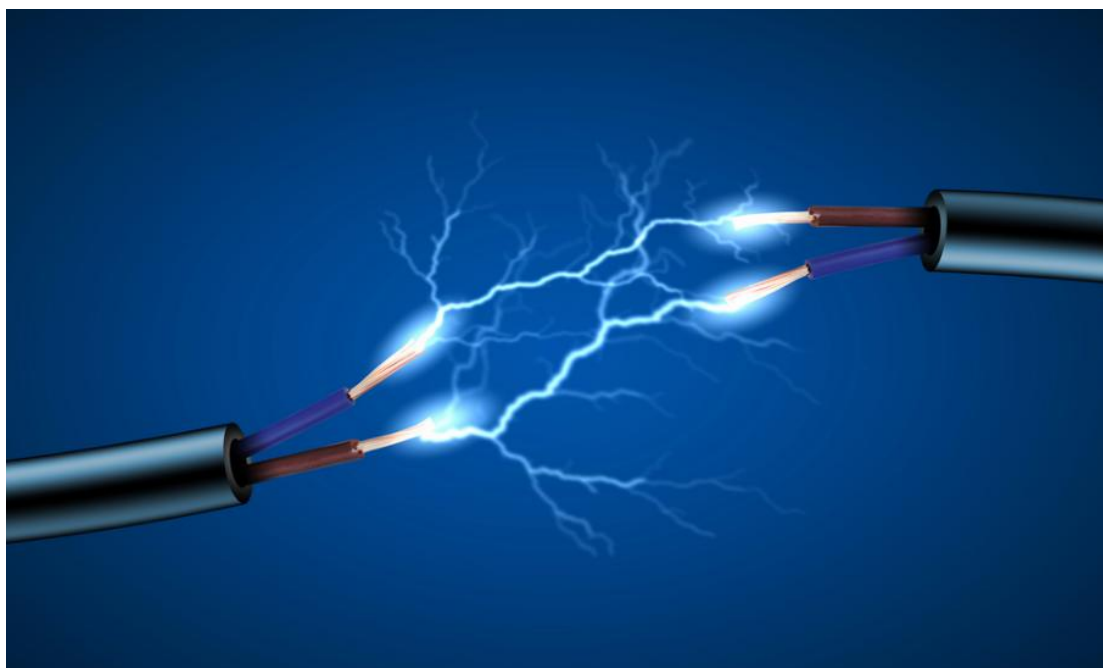
经过一系列的改进和创新，蒸汽机得以逐步改变世界。托马斯·纽科门（Thomas Newcomen）发明出了利用蒸汽机更好地从矿井中抽水的方法；詹姆斯·瓦特（James Watt）和马修·博尔顿（Matthew Bolton）则使蒸汽动力能更有效地循环运动；之后，高压蒸汽机的出现催生了蒸汽机车的广泛应用，极大地便利了生产和运输。



[illegible]

Unit 4

Electricity



Looking Back



Reading 1:

Who Invented Electricity? -
Energy, Power and History



Reading Comprehension



Skills: Delivery



Looking Forward



Reading 2:

What Will the Batteries of the
Future Be Like?



Translation



Further Discussion



Looking Back

Who Invented Electricity? - Energy, Power and History³

1. When it comes to electricity invention or the discovery of energy power, it is actually a long story, over a long time. Electrical light is so common to this modern world in the form of lightning, but the question of who invented electricity should actually be what was involved in discovering electricity. Electricity was common in the natural world, but electric power was the result of experiment.
2. From the writings of Thales of Miletus it appears that Westerners knew as long ago as 600 B.C. that amber becomes charged by rubbing. There was little real progress until the English scientist William Gilbert in 1600 described the electrification of many substances and coined the term electricity from the Greek word for amber. As a result, Gilbert is called the father of modern electricity. In 1660 Otto von Guericke invented a crude machine for producing static electricity. It was a ball of sulfur, rotated by a crank with one hand and rubbed with the other.
3. Progress quickened after the Leyden jar was invented in 1745 by Pieter van Musschenbroek. The Leyden jar stored static electricity, which could be discharged all at once. In 1747 William Watson discharged a Leyden jar through a circuit, and comprehension of the current and circuit started a new field of experimentation.
4. A new interest in electric current began with the invention of the battery. Luigi Galvani had noticed (1786) that a discharge of static electricity made a frog's leg jerk. Consequent experimentation produced what was a simple electron cell using the fluids of the leg as an electrolyte and the muscle as a circuit and indicator. Galvani thought the leg supplied electricity, but Alessandro Volta thought otherwise, and he built the voltaic pile, an early type of battery, as proof.

³ The original article is available at <https://www.electricityforum.com/who-invented-electricity>

A Different Kind of Power: The Battery

5. Learning how to produce and use electricity was not easy. For a long time, there was no dependable source of electricity for experiments. Finally, in 1800, Alessandro Volta, an Italian scientist, made a great discovery. He soaked paper in salt water, placed zinc and copper on opposite sides of the paper, and watched the chemical reaction produce an electric current. Volta had created the first electric cell.

6. By connecting many of these cells together, Volta was able to “string a current” and create a battery. It is in honor of Volta that we rate batteries in volts. Finally, a safe and dependable source of electricity was available, making it easy for scientists to study electricity.

A Current Began

7. An English scientist, Michael Faraday, was the first one to realize that an electric current could be produced by passing a magnet through a copper wire. It was an amazing discovery. Almost all the electricity we use today is made with magnets and coils of copper wire in giant power plants. Both the electric generator and electric motor are based on this principle. A generator converts motion energy into electricity. A motor converts electrical energy into motion energy.

8. In 1819 Hans Christian Oersted discovered that a magnetic field surrounds a current-carrying wire. Within two years André Marie Ampère had put several electromagnetic laws into mathematical form, D. F. Arago had invented the electromagnet, and Michael Faraday had devised a crude form of electric motor. Practical application of a motor had to wait 10 years, however, until Faraday (and earlier, independently, Joseph Henry) invented the electric generator with which to power the motor. A year after Faraday's laboratory approximation of the generator, Hippolyte Pixii constructed a hand-driven model. From then on engineers took over from the scientists, and a slow development followed; the first power stations were

built 50 years later.

9. In 1873 James Clerk Maxwell had started a different path of development with equations that described the electromagnetic field, and he predicted the existence of electromagnetic waves traveling with the speed of light. Heinrich R. Hertz confirmed this prediction experimentally, and Marconi first made use of these waves in developing radio (1895).

Mr. Edison and His Light

10. Thomas Alva Edison is one of the greatest of all inventors and is normally credited with creating light bulbs (along with Nicolas Tesla). He arrived in Boston in 1868. In Boston he found men who knew something of electric current, and, as he worked at night and cut short his sleeping hours, he found time for study. He bought and studied Faraday's works. Presently came the first of his multitudinous inventions, an automatic vote recorder, for which he received a patent in 1868.

11. In 1879, Thomas Edison focused on inventing a practical light bulb, one that would last a long time before burning out. The problem was finding a strong material for the filament, the small wire inside the bulb that conducts electricity. Finally, Edison used ordinary cotton thread that had been soaked in carbon. This filament didn't burn at all—it became incandescent; that is, it glowed.

12. The next challenge was developing an electrical system that could provide people with a practical source of energy to power these new lights. Edison wanted a way to make electricity both practical and inexpensive. He designed and built the first electric power plant that was able to produce electricity and carry it to people's homes.

13. Edison's Pearl Street Power Station started up its generator on September 4, 1882, in New York City. About 85 customers in lower Manhattan received enough power to light 5,000 lamps. His customers paid a lot for their electricity, though. In

Unit 4 Electricity—Looking Back

today's dollars, the electricity cost \$5.00 per kilowatt-hour! Today, electricity costs about 12.7 cents per kilowatt-hour for residential customers, about 11 cents for commercial, and about 7 cents per kilowatt-hour for industry.

The Question: AC or DC?

14. The turning point of the electric age came a few years later with the development of AC (alternating current) power systems. Croatian scientist, Nikola Tesla came to the United States to work with Thomas Edison. After a falling out, Tesla discovered the rotating magnetic field and created the alternating current electrical system that is used very widely today. Tesla teamed up with engineer and business man George Westinghouse to patent the AC system and provide the nation with power that could travel long distances – a direct competition with Thomas Edison's DC system. Tesla later went on to form the Tesla Electric Company, invent the Tesla Coil, which is still used in science labs and in radio technology today, and design the system used to generate electricity at Niagara Falls.

15. Now using AC, power plants could transport electricity much farther than before. While Edison's DC (direct current) plant could only transport electricity within one square mile of his Pearl Street Power Station, the Niagara Falls plant was able to transport electricity over 200 miles!

16. Electricity didn't have an easy beginning. While many people were thrilled with all the new inventions, some people were afraid of electricity and wary of bringing it into their homes. They were afraid to let their children near this strange new power source. Many social critics of the day saw electricity as an end to a simpler, less hectic way of life. Poets commented that electric lights were less romantic than gaslights. Perhaps they were right, but the new electric age could not be dimmed.

17. In 1920, about two percent of U.S. energy was used to make electricity. In 2017, with the increasing use of technologies powered by electricity, it was almost 40 per cent.

Reading Comprehension

Part A: Generalizing and Summarizing

Choose the correct name to fill the blanks.

A. Michael Faraday

B. William Gilbert

C. D. F. Arago

D. William Watson

E. Heinrich R. Hertz

F. André Marie Ampère

G. Guglielmo Marconi

H. Luigi Galvani

I. George Westinghouse

The discovery and development of electricity involved many individuals and milestones over centuries. As early as 600 B.C., Thales of Miletus observed that amber became charged when rubbed. 1. _____ popularized the term "electricity" and is considered the father of modern electricity. Otto von Guericke, in 1660, created a device to produce static electricity. Progress accelerated with the invention of the Leyden jar in 1745 by Pieter van Musschenbroek, which stored static electricity. 2. _____ conducted an experiment in 1786 that led Alessandro Volta to create the first battery in 1800, providing a reliable source of electric power. 3. _____ made significant contributions by discovering that an electric current could be generated by moving a magnet through a wire coil, leading to the development of generators and motors. Subsequent advancements in electromagnetism were made by Hans Christian Oersted, 4. _____, and D.F. Arago. Thomas Edison, celebrated for inventing the practical light bulb in 1879, also built the first electric power plant in 1882. His direct current (DC) system competed with Nikola Tesla's and George Westinghouse's alternating current (AC) system, which proved superior for long-distance electricity transmission.

Part B: Paraphrasing and Creating

Paraphrase the underlined sentences in paragraph 16 in your own words.

Unit 4 Electricity—Looking Back

“Electricity didn’t have an easy beginning. While many people were thrilled with all the new inventions, some people were afraid of electricity and wary of bringing it into their homes. They were afraid to let their children near this strange new power source. Many social critics of the day saw electricity as an end to a simpler, less hectic way of life. Poets commented that electric lights were less romantic than gaslights. Perhaps they were right, but the new electric age could not be dimmed.”

Part C: Analyzing and Extracting

Choose the appropriate letters A-D for each question.

6. Which of the following statements best explains why William Gilbert is known as the father of modern electricity? _____

- A. He invented the first electric generator.
- B. He described the electrification and coined the term electricity.
- C. He discovered the relationship between electricity and magnetism.
- D. He built the first electric cell using a frog's leg as an electrolyte.

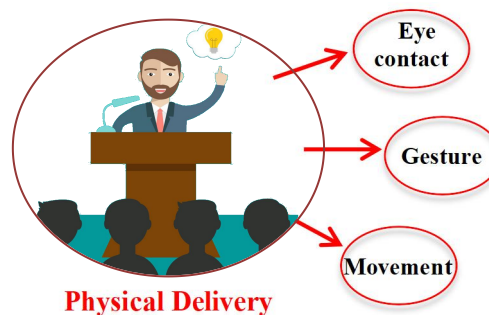
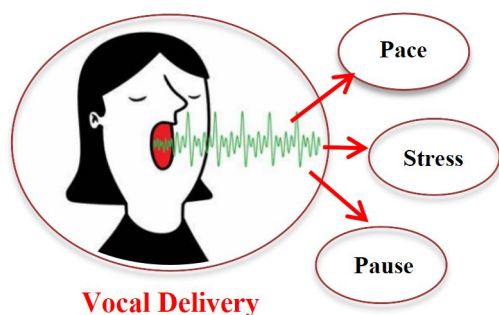
7. What was the significant contribution of Alessandro Volta to the field of electricity? _____

- A. He invented the first static electricity machine.
- B. He discovered that a magnetic field surrounds a current-carrying wire.
- C. He created the first electric cell and an early type of battery.
- D. He invented the first electric motor.

8. What was a major limitation of Thomas Edison's DC system compared to the AC system developed by Nikola Tesla?
- A. Edison's DC system could only transport electricity within a limited range.
 - B. Edison's DC system was more expensive to produce.
 - C. Edison's DC system could not be used for lighting.
 - D. Edison's DC system was less safe for home use.

Skills

Delivery

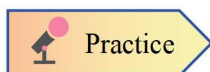


Vocal Delivery (Voice)		Physical Delivery (Body)	
Pace	➤ Native speakers: 500 words in 3 minutes (167 wpm) ➤ Most: 400 words in 3 minutes (133 wpm). ➤ Non-native: about <u>120</u> words per minute.	Eye contact	➤ Look at your audience <u>personally and pleasantly</u>; ➤ <u>Shift</u> your gaze every few sentences; ➤ Alternatively, choose <u>4-5</u> <u>people</u> to look at.
Stress	Use stress to <u>clarify</u> things and <u>emphasize</u> key points.	Gesture	Use gestures to <u>convey your meaning</u> and make you appear <u>natural and confident</u> as well.
Pause	Leave a pause after a particular word or sentence to <u>add extra emphasis</u> to your speech or create a <u>dramatic impact</u> on the audience.	Movement	Two types of movements: ➤ Repeatedly shift the weight from one foot to the other; ➤ Pace back and forth across the podium.
Tips for using voice and body as tools: ✧ Slow down when speaking; ✧ Pause after key information; ✧ Use filler words when necessary; ✧ Maintain eye contact with the audience; ✧ Use appropriate gestures; ✧ Stand naturally on the stage.			

Sample for delivery:

Q: How is electricity generated in a thermal power station?

Electricity generation in a thermal power station involves converting chemical energy into electrical energy through a series of steps. First, coal, oil, or gas is burned in a boiler, releasing heat and converting chemical energy into thermal energy. Second, the heat generated boils water into steam, which is then used to rotate a turbine. This process converts thermal energy into mechanical energy. Third, the turbine's rotation drives a generator, which produces electricity through electromagnetic induction. In conclusion, the thermal power station efficiently converts fuel's chemical energy into electrical energy, powering homes and industries.



Select a topic from the list and deliver a one-minute presentation. Pay attention to your vocal delivery and physical delivery.

1. Advantages of AC over DC

2. Possible methods to save electricity in daily life



Looking Forward

What Will the Batteries of the Future Be Like?⁴

By Por Beatriz Guillén

1. In recent years, tech magnate Elon Musk has promised to conquer space, make it possible to travel from Los Angeles to San Francisco in just 35 minutes thanks to vacuum tube transport, and fill the roads with electric and self-driving cars, among other ambitious schemes. Some of these projects are still alive and well; others, not so much. But the future to which the founder of PayPal, Tesla and SpaceX aspires involves massive energy needs that today more than ever call for new, more capable and environmentally friendly solutions. This is a demand common to every aspect of the technological push in which we are immersed: from a small smartphone to an enormous solar power plant, we need new energy storage systems, and that is why batteries have become the great obsession of technology companies. The search is on for the battery of the future: more powerful, longer-lasting, safer and more sustainable.

2. In our technological age, lithium-ion batteries have become the common industry standard, used in everything from mobile phones to electric vehicles, and have made all our portable devices possible. But they have their limitations. One of them, and one that has caused quite a stir, relates to safety. In September 2016, a Samsung Galaxy Note 7 exploded and set fire to the Jeep in which it was charging. Less than a month later, a commercial airliner had to be evacuated when a phone of the same model started to smoke. By then, dozens of cases of burning handsets had already been reported and the South Korean company had halted production. Today we know that the cause of these explosions was caused by faulty battery design, and that the problem has also affected other devices. And although these are anecdotal and very rare cases, they have caused great concern among

⁴ The original article is available at <https://www.bbvaopenmind.com/en/technology/future/what-will-the-batteries-of-the-future-be-like/>

users.

3. The explosion of batteries highlights a challenge: shrinking the size of batteries and at the same time providing them with more capacity carries great risks. For this reason, companies have now taken a “think-outside-the-box” approach to the problem and changed the direction of their efforts. Today, the near future for batteries is not that they last longer, but that they can be charged much faster. The goal is to get hours of use from just a few minutes of charging.



(A Samsung Note 7 after its explosion caused by faulty battery design.)

Companies and Universities in Search of the Golden Battery

4. In recent years, companies and researchers have focused on fast charging. Qualcomm, the company that manufactures the “brains” of a large number of devices, has released successive versions of its Quick Charge charging system. The latest to date, version 5, promises a 50% charge in five minutes with 70% more efficiency. Samsung and Huawei offer their Super Fast Charging and Supercharge systems, respectively, with the same intent.

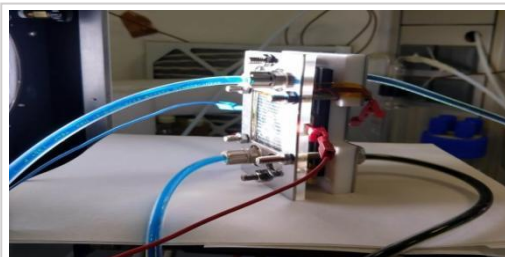
5. Meanwhile, researchers are exploring new ways to speed up the charging process, which depends on the speed of the transport of lithium inside the electrodes. This in turn is influenced by multiple factors including the battery cell design and the materials of the anode, cathode and electrolyte. Some research has found methods to speed up the movement of lithium, for example by adding large metal ions such as potassium. Faster charging is of particular interest for electric vehicles, whose use is limited by recharging time. However, fast charging systems have the disadvantage that lithium tends to be more easily immobilised between the carbon layers of graphite on the anode, which in turn degrades this

electrode. This results in a loss of battery function.

Counteracting Degradation

6. This progressive deterioration is another major problem with today's lithium-ion batteries. Repeated charging, especially in mobile phones, leads to rapid degradation and a loss of some of their effectiveness after about a year of use. The lifespan of batteries is about 1,000 complete charge/discharge cycles, and so far no possibility of extending this has been found. Scientists are studying methods and structures of new batteries to extend the lifetime by facilitating fast charging while avoiding lithium deposition at the anode. They are also exploring possible new avenues such as replacing batteries with flexible or cable-like supercapacitors that promise a full charge in seconds and a lifetime of 30,000 charge/discharge cycles, and would take up much less space than today's batteries.

7. On the other hand, new proposals include different solutions for different uses. One example is flow batteries that store energy in external liquid tanks, which could last more than a decade without degrading. One such battery developed at Harvard University uses



(Solid-state batteries achieve greater energy storage with longer life and greater safety with less toxicity.)

organic molecules dissolved in pH-neutral water, a formulation so stable that the batteries will lose only 1% of their capacity for every 1,000 cycles. They are also considerably safer, as this solution contains no corrosive or toxic elements. Such batteries are suitable for storing energy produced by solar or wind power plants, but could also be adapted for use in households.

8. Such batteries could help mitigate another problem with existing batteries: their ecological impact. In addition to using toxic and corrosive compounds, lithium-ion batteries have such a high environmental cost—mainly due to the extraction of the metals and rare earths used in their production—that manufacturing an electric vehicle has a larger pollution and emissions footprint than a conventional one.

Unit 4 Electricity—Looking Forward

However, this is offset over its lifetime.

9. But on the road to greener batteries, a current trend is to explore solid-state batteries, which replace liquid or gel electrolyte with solid electrolyte, achieving greater energy storage with longer life and greater safety with less toxicity, but with the trade-off of slower charging. This technology is currently being applied to lithium-ion batteries and is already being considered for commercial use in automobiles, although at the moment the price is high.

10. In short, there are many and varied avenues being pursued in the quest for new batteries that are more capable, faster, safer and more sustainable, using alternative materials such as manganese, sodium, silicon or vanadium, dispensing with other increasingly expensive and scarce materials such as cobalt (present in lithium-ion batteries), and applying designs based on nanotechnology. At the moment, the vast majority of these proposals are still in the research, development or experimental use phase, and there is as yet no clear winner that promises to make our old, familiar lithium-ion batteries obsolete. Thus it seems likely that we will still be dependent on them for the foreseeable future. But with experts warning of an expected lithium shortage in just a few years, the batteries of the future will need to get charged up to quickly jump from the lab to our devices.

Translation

反复充放电会导致锂离子电池在使用一年后快速退化，所以科研人员正在研究如何延长其使用寿命。哈佛大学的研究团队开发了一种液流电池，他们将有机分子溶解在 PH 值中性的水中，使得锂离子电池充放电 1000 次后容量仅损失 1%。但这种电池离商业化应用还有一段很长的路要走。



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